

```
#include <stdio.h>
#include <stdlib.h>

#define MAX 5

int queue[MAX];
int front = -1;
int rear = -1;

void enqueue(int val) {
    if (rear == MAX - 1) {
        printf("\nQueue Overflow!\n");
    } else {
        if (front == -1) front = 0;
        rear++;
        queue[rear] = val;
        printf("Successfully inserted %d\n", val);
    }
}

void dequeue() {
    if (front == -1 || front > rear) {
        printf("\nQueue Underflow!\n");
    } else {
        printf("\nDeleted element: %d\n", queue[front]);
        front++;
        if (front > rear) {
            front = -1;
            rear = -1;
        }
    }
}

void display() {
    if (front == -1) {
        printf("\nQueue is empty!\n");
    } else {
        printf("\nQueue elements: ");
        for (int i = front; i <= rear; i++) {
            printf("%d ", queue[i]);
        }
        printf("\n");
    }
}

int main() {
    int choice, val;
    while (1) {
        printf("\n--- LINEAR QUEUE MENU ---\n");
        printf("1. Enqueue  2. Dequeue  3. Display  4. Exit\n");
        printf("Enter choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                printf("Enter value to insert: ");
                scanf("%d", &val);
```

```
--- LINEAR QUEUE MENU ---
1. Enqueue  2. Dequeue  3. Display  4. Exit
Enter choice: 1
Enter value to insert: 10
Successfully inserted 10

--- LINEAR QUEUE MENU ---
1. Enqueue  2. Dequeue  3. Display  4. Exit
Enter choice: 1
Enter value to insert: 20
Successfully inserted 20

--- LINEAR QUEUE MENU ---
1. Enqueue  2. Dequeue  3. Display  4. Exit
Enter choice: 1
Enter value to insert: 30
Successfully inserted 30

--- LINEAR QUEUE MENU ---
1. Enqueue  2. Dequeue  3. Display  4. Exit
Enter choice: 3

Queue elements: 10 20 30

--- LINEAR QUEUE MENU ---
1. Enqueue  2. Dequeue  3. Display  4. Exit
Enter choice: 2

Deleted element: 10

--- LINEAR QUEUE MENU ---
1. Enqueue  2. Dequeue  3. Display  4. Exit
Enter choice: 3

Queue elements: 20 30

--- LINEAR QUEUE MENU ---
1. Enqueue  2. Dequeue  3. Display  4. Exit
Enter choice: 4
Exiting program...
```